

MESHACH ANDO

Jacksonville, FL | · 904-749-2799 | meshachando.apply@gmail.com | linkedin.com/in/meshachando

EDUCATION

University of North Florida (UNF)

MSc Civil Engineering (Focus: Environmental/Solid Waste)

Jacksonville, FL

Aug 2024 – Present

- **CGPA:** 3.96/4.00 | **Expected Graduation:** May 2026
- **Research Focus:** Advanced computational modeling of landfill methane emissions, PFAS dispersion, and leachate optimization.

Kwame Nkrumah University of Science and Technology (KNUST)

BSc Civil Engineering (First Class Honors).

Kumasi, Ghana

Aug 2018 – Nov 2022

TECHNICAL SKILLS

- **Engineering & Geospatial Software:** AutoCAD, Civil 3D, ArcGIS Pro, ArcPy Automation, Spatial Analyst Tools.
- **Modeling & Analysis:** Python (Pandas, NumPy, SciPy, Matplotlib), Genetic Algorithms, Gaussian Dispersion Modeling.
- **Field Methodologies:** Surface Emissions Monitoring (SEM), Waste Characterization (ASTM standards), Leachate Sampling.

CERTIFICATIONS

- Fundamentals of Engineering (FE) – Civil, passed February 2026.

GRADUATE RESEARCH EXPERIENCE

University of North Florida

Graduate Research Assistant (De la Cruz Research Lab)

Jacksonville, FL

Aug 2024 – Present

- Developed a Python-based inverse modeling framework using Genetic Algorithms to quantify whole-landfill methane emissions. Optimized source location (x, y) and emission rates (Q) using SEM concentration data, reducing computational runtime by ~60 minutes per unit landfill iteration compared to industry-model.
- Spearheaded a comparative analysis of 15 years of temporal data (2010–2025) across four major U.S. landfills. Identified an overestimation in inventory models (IPCC, EPA GHGRP, SWICS, CARB, Dutch Afvalzorg) when compared to Surface Emissions Monitoring (SEM) and satellite measurements.
- Developed a Gaussian dispersion model to map PFAS attenuation from area sources. Calculated population exposure risks at 1km, 3km, and 6km radii to support regulatory buffer zone recommendations.
- Programmed an end-user estimation tool (Python/Excel integration) for landfill operators to calculate site-specific carbon footprints, automating data intake from operational logs.

PROFESSIONAL EXPERIENCE

Kumasi Compost and Recycling Plant (KCARP)

Research Assistant / Process Engineer (National Service)

Kumasi, Ghana

Nov 2022 – Jan 2024

- Monitored the throughput of a 2,400 TPD Integrated Solid Waste Management Facility (ISWMF). Utilized material flow analysis (MFA) characterization to compare waste inflow vs. recovered fractions, identifying bottlenecks in the sorting lines.
- Executed daily physicochemical profiling of compost piles (C:N ratio, moisture, temperature) to optimize aerobic degradation rates and ensure compliance with agricultural quality standards.
- Managed flow-rate monitoring and heavy metal analysis for the leachate treatment plant, ensuring effluent met environmental discharge limits.
- Led field characterization studies to determine waste composition, directly informing machine calibration for higher recovery rates of recyclables.

Ghana Highway Authority (Materials Division)

Accra, Ghana

Engineering Intern

Jun 2021 – Sep 2021

- Conducted geotechnical soil testing (Atterberg limits, sieve analysis, compaction) relevant to earthworks and potential landfill liner specifications.

Zoomlion Company Limited

Obuasi, Ghana

Operations Intern

Jun 2019 – Sep 2019

- Audited daily operations at the engineering landfill and waste stabilization ponds, tracking cell compaction and cover application compliance.
- Compiled and digitized environmental monitoring data for regional reporting, tracking collection fleet efficiency and disposal volumes.

PROFESSIONAL AFFILIATIONS

- Student Member, American Society of Civil Engineers (ASCE)
- Student Member, National Society of Black Engineers (NSBE)
- Student Member, Institute of Transportation Engineers (ITE/ITS)